2025 CSC493 – Capstone Weekly Reports[1]

<Download this template and name it Week_<WEEK_NUMBER>, where "<REPORT_NUMBER>" is replaced by the current week number (i.e., "1" for the first weekly report and so forth).

Week	10
Project Name	Moodr AI
Report Date	Oct 28, 2025 ← click to select date

Part 1: Weekly Progress Report

- **Accomplishments**: **What did you complete this week?** Be specific and measurable.
- Examples: *Implemented login authentication using OAuth2; Fixed bug in file upload function (issues #14); Completed draft of UI wireframes; Identified functional and non-functional requirements; etc. (up to 3 points)*

I have fixed issues with the CNN model misclassifying songs into either just happy or sad moods due to overconfidence, and have connected the CNN model to the GUI allowing users to choose between scikit ML models and the CNN model.

- **Challenges**: **What problems did you encounter?** Mention blockers, unresolved issues, or risks.
- Examples: *Struggled with connecting database migrations to production server; API documentation was incomplete – need clarification; etc. (up to 3 points)*

The problem that I ran into that took the most time to fix this week was that of overconfidence by the CNN model, my CNN model was classifying most songs strongly towards sad or happy moods despite certain ones I tested for being angry and calming songs. To fix this issue I had to implement global feats to be extracted along with the spectrogram for the CNN model to use for classification. This seems to have fixed the problem.

- **Next Steps**: **What are your goals for next week?** Keep them realistic and linked to project milestones.
- Examples: Set up automated testing with GitHub Actions; Final project proposal *(up to 2 points)*

My goals for next week are to adjust the CNN model's class weighting for better accuracy, implement data augmentation of the dataset such as pitch-shifting, time-stretching, and random cropping for better accuracy on lower quality music files, and updating GUI to show more results about accuracy, probability, and confidence of results.

- **Discussion Points**: **What would you like to discuss in class, or with your professor/TA?** This could include technical issues, project direction, or collaboration concerns.
- Examples: *Should we prioritize front-end features before database scaling? Need guidance on writing integration tests; etc. (up to 2 points)*

Should I continue refining the CNN architecture this week or shift focus to report preparation and GUI polish? My GUI successfully loads both Scikit and CNN models, but should I include a confidence visualization for clarity in user testing?

[1] Detailed Weekly Report requirements can be found here: 2024 URCPP Capstone - Using Agile and Reporting Out

Part 2: Time Reporting

Make sure that as you fill out the first prompt, you include in enough detail in the summary. For example, "debugging" is vague, but "debugged function X to make sure that when user does action Y, it is called and returns the value Z" is better.

- **Time Spent**: Briefly explain how much time, *outside of class*, spent on your project. If you worked on multiple components, each should get a detailed summary. Make sure to add up all the hours and minutes correctly. *Add as many rows as you need to the table below. Please do not include hours in class as part of this section. (up to 4 points)*

START	FINISH	HOURS	DETAILED SUMMARY
MM/DD HH:MM	MM/DD HH:MM		
10/21 08:00	10/21 14:00	6	I spent 6 hours on Tuesday mostly debugging the CNN model as I noticed that using test songs of the 4 different moods I am classifying, most of the moods tended to group around sad and happy despite certain songs clearly being angry / calm songs. I did not manage to fully debug that issue on Tuesday alone.
10/22 15:00	10/22 19:00	4	I started debugging the issue of the grouping around mostly happy and sad songs at first, then at around 2 hours in, I decided to add the cnn model to the GUI and remove the imported theme I was aiming to use and replace it with a simple purple, grey, and light grey color scheme.
10/23 08:00	10/23 14:00	6	Fully debugged the CNN model's issue and "solved" the overconfidence that the model was having by adding global feats in addition to spectrograms to the model for classification purposes. Ultimately, the model seems to now classify more accurately, I am not 100% sure that the calm and angry mood ranges are still accurate enough, but songs within those moods are falling into that range now.
	WEEKLY TOTAL	14	

- **Total (Cumulative) Project Time Spent**: After the number of hours and minutes, make sure to briefly explain whether you are on track and if not, what you may need to do in order to achieve what you set out to accomplish. *(up to 2 points)*

I have spent 136 hours total on this project so far. I am currently on pace with the timeline for the project and technically am slightly ahead as I was supposed to test my global features addition this week (October 27-Nov 2), but I opted to test it ahead of time to truly see if that fixed the problem of over confidence from the CNN model.

Rubric:

The following rubric will be used, but they might change as needed.

Accomplishments (3 points)

1 point for a general description of progress, 2 points for specifics on progress, 3 points for specifics AND referring to previous targets and explaining how current accomplishments build on previous ones.

Challenges (3 points)

1 point for mentioning there are roadblocks, 2 points for specifics, 3 points for specifics AND what was done already to try to overcome them.

Desired discussion points (2 points)

1 point for at least one relevant discussion point as a general question, 2 points for relevant discussion points with specifics

Future Goals (2 points)

1 point for concrete future targets (i.e. "working more on the project" is a zero, but "working on getting component X to interface with component Y" suffices), 2 points for tying in the targets with what was hopefully discussed in the meeting.

Time Spent (4 points)

1 point for including general statements of how much time was spent ("4 hours on coding"), 2 points for splitting time into specific parts ("1.5 hours on research on component X, 1 hour coding, 2.5 hours debugging"), 3 points for specific parts and details on the pieces ("1.5 hours researching Turtle interface for drawing concentric circles given inputs from the user, 1 hour coding function X that used that interface, 2.5 hours testing function X by giving it multiple values and fixing errors for values A, B, C, and D"). 1 Point for totalling the hours correctly.

What happens if your time on a task is interrupted and you don't have a concrete (or discrete) end time? In this case put the start time in, and the word "interrupted" for the end time and include the task total time. Rounding to the nearest 15 minutes is acceptable. *(This makes adding up times easier, especially when you use decimal hours, i.e. 3.75 rather than 3 hours 45 minutes.)*

Total (Cumulative) Project Time (2 points)

1 point for summing the values correctly, 2 points for the total time AND reflection on progress (you are confident to fit the target and if not, what course corrections you anticipate needing to make)

Resources

Here's a link to this Weekly Report Template – Make a copy and use it:

2025 CSC493 Weekly Report v2 TEMPLATE